

Moving from analyzing observational studies that
use random sampling to analyzing experiments
that use random assignment

MATH 213

Two questions from last week:

Analyzing observational studies that use random sampling

Last week, we spent time analyzing observational studies that use random sampling. Because random sampling (essentially) was used, we can generalize from the sample to the population.

The two questions were:

1. What is the difference between the proportion of males who donated blood and the proportion of females who donated blood in the United States in 2004?
1. What is the difference in average age at first marriage between those US residents in 1972 and US residents in 2006?

In both cases, we used a sample statistic and a confidence interval to be able to estimate a population parameter.

Donating blood:

Samples: 1271 male respondents, 1427 female respondents to the GSS in 2004

Populations of interest: the populations of all male US residents and all female US residents

Study type: Observational – we cannot infer causation.

Random sampling? YES – we can generalize from the sample to the population of interest

Sample statistic: the difference in the proportion of males from the sample who gave blood and the proportion of females in the sample who gave blood

Use this to estimate the population parameter: the difference in the proportion of males from the population who gave blood and the proportion of females from the population who gave blood. I.e. the percentage point difference between proportions of males and females who donated blood in the United States in 2004

Age at first marriage:

Samples: the 1395 respondents to the GSS in 1972, the 1161 respondents to the GSS in 2006

Populations of interest: the population of all US residents who had been married at least once in 1972 and the population of all US residents who had been married at least once in 2006

Study type: Observational – we cannot infer causation.

Random sampling? YES – we can generalize from the sample to the population of interest

Sample statistic: the difference in sample mean age at first marriage

Use this to estimate the population parameter: the difference in population mean age at first marriage. I.e. how much older, on average, people were when they got married for the first time among US residents in 2006 compared to 1972

Today: Analyzing Experiments with Random Assignment (rather than Observational Studies that use Random Sampling)

Back to:

1. Bank supervisor experiment
2. Weight loss experiment

Sex discrimination case study

Because the experiment is blind and each manager is given only ONE file (M or F) to review, we only have information on what each manager would do in ONE of these situations. Here is what we ACTUALLY know:

Manager	Male or female file	Decision	Action if M file	Action if F file
1	M	promotion	promotion	
2	M	no promotion	no promotion	
3	M	promotion	promotion	
4	F	no promotion		no promotion
5	F	no promotion		no promotion
6	F	promotion		promotion

What parameter can we estimate?

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6	F	promotion		promotion

What parameter can we estimate?

The difference in proportions of those who would promote if all study participants were given a male file and if all study participants were given a female file.

Weight Loss Experiment

What we know:

Group	Weight loss at 16 weeks (pounds)	Weight loss with incentive	Weight loss with control
Incentive	24	24	
Control	-17		-17
Incentive	15	15	
Incentive	25	25	
Control	-2		-2
Incentive	6	6	

What parameter can we estimate?

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What parameter can we estimate?

How much more weight, on average, people would lose on the incentive plan (if ALL participants were on the incentive plan) than on the control plan (if ALL participants were on the control plan).